

Note : In this article we have studied Maxwell's equations for static field. In practice field always vary with time. Maxwell's equations for this field has to be written separately, these are not same for static fields. Only Gauss law, for electric and magnetic is so powerful that it does not require any modification. All other laws require modification.

8.3 Faraday's Law :

MU - May 2010

In 1820, Oersted demonstrated that when a current is passed through filament it deflects the compass needle.

It proved that a current is producing the magnetic field. Then Faraday thought that if current could produce a magnetic field, then a magnetic field should be able to produce a current.

He worked on this problem over a period of 10 years. In 1831 he proved it experimentally. Refer Fig. 8.3.1.

- He wound two separate windings on the iron toroid.
- A galvanometer was placed in one circuit and a battery in the other.
- Upon closing the switch in the battery circuit he noted a momentary deflection of the galvanometer.
- When the switch was open, the galvanometer was again deflected momentarily in the opposite direction.
- The deflection of the galvanometer proved that a current is produced in the galvanometer circuit.

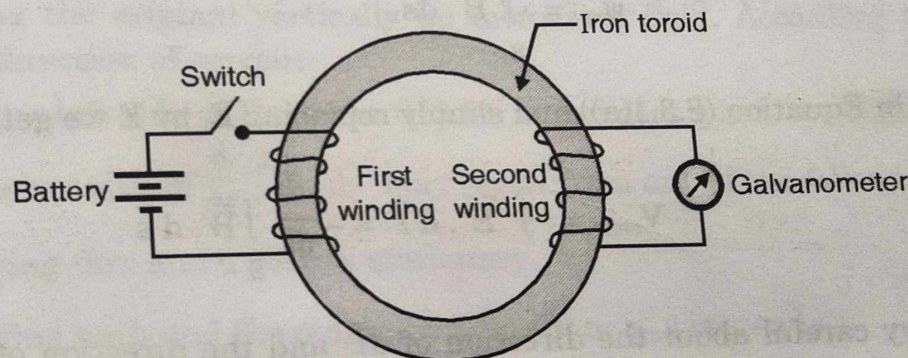


Fig. 8.3.1 : Faraday's experiment

The current is produced in the second winding whenever the switch is closed or opened. What happens is when the switch is closed the current flows through the first coil producing changing magnetic field, which induces an emf (V_{emf} measured in volts) which is responsible for current in the galvanometer circuit. Similar action takes place at the time of opening the switch. To flow a current in the galvanometer circuit, an emf (called an electromotive force) is required to be generated in the second circuit. Thus Faraday's discovery was :

The electromotive force, V_{emf} around a closed circuit is equal to the negative of time rate of change of magnetic flux enclosed by that path.

Mathematically,

$$V_{\text{emf}} = \oint \vec{E}_f \cdot d\vec{l} = - \frac{d\psi_m}{dt} \quad \dots(8.3.1(a))$$

If there are N number of turns in that circuit, then

$$V_{\text{emf}} = \oint \vec{E}_f \cdot d\vec{l} = - N \frac{d\psi_m}{dt} \quad \dots(8.3.1(b))$$

where ψ_m = the magnetic flux through each turn.

N = number of turns.

E_f = induced electric field in the closed circuit which is the cause of emf in that circuit.

Note the difference between the electric field (E_f) and the field due to charge (also called electrostatic field) E_e .

1. The electrostatic field (E_e) is conservative (as we have studied in second chapter), giving,

$$\oint \vec{E}_e \cdot d\vec{l} = 0 \quad \dots(8.3.2(a))$$

2. But the emf produced field (E_f) is non conservative

$$\oint \vec{E}_f \cdot d\vec{l} \neq 0 \quad \dots(8.3.2(b))$$

The flux can be obtained from flux density using

$$\psi_m = \int_s \vec{B} \cdot d\vec{s}$$

Introducing this in Equation (8.3.1(a)) and simply replacing E_f by E we get,

$$V_{\text{emf}} = \oint \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_s \vec{B} \cdot d\vec{s} \quad \dots(8.3.3)$$

One should be very careful about the direction of $d\vec{l}$ and the direction of $d\vec{s}$. These are related using right hand, where the fingers of our right hand indicate the direction of the closed path, and our thumb indicates the direction of $d\vec{s}$. A flux density $\vec{B}$ in the direction of $d\vec{s}$ and increasing with time (d/dt of it is positive) thus produces an average value of $\vec{E}$ which is opposite (due to negative sign on right hand side) to the positive direction about the closed path.

8.3.1 Lenz's Law :

The negative sign on the right hand side shows that the induced voltage acts in such a way as to oppose the cause producing it (i.e. ψ_m or $\vec{B}$). This is called as Lenz's law.

Electromagnetic Induction Time Varying Fields & Maxwell's Equations

In other words, the current flows in the circuit in such a direction that the magnetic field produced by this current will oppose the original magnetic field. To understand the concept consider a simple situation as shown in Fig. 8.3.2.

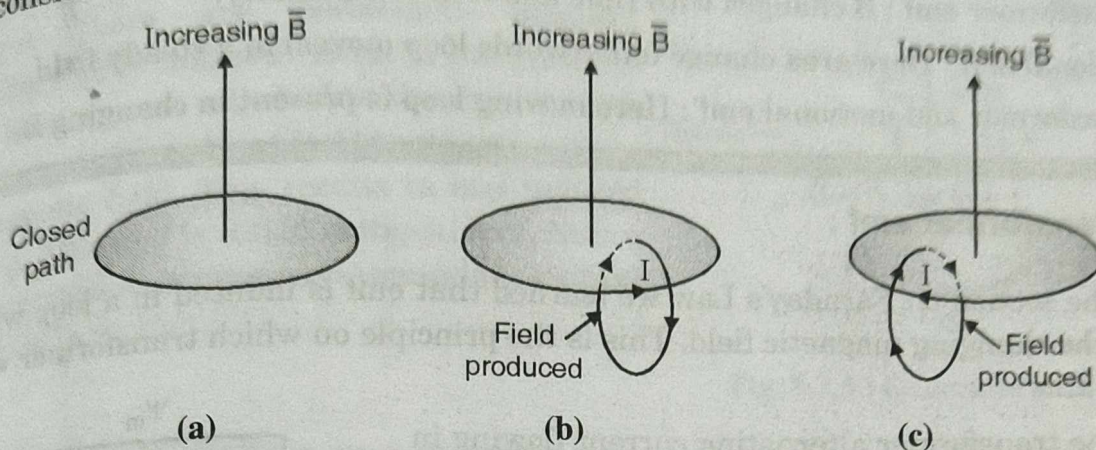


Fig. 8.3.2

In Fig. 8.3.2(a), the flux density (or flux) is assumed increasing vertically, which induces emf (V_{emf}) in the closed path resulting in current I in that path.

This current in turn produces a magnetic field. The relation between the direction of current and the magnetic field is obtained using right hand thumb rule. Which states that if we grip the current filament in our right hand such that the direction of thumb is in the direction of current then the direction of fingers indicate the direction of field.

In Fig. 8.3.2(b), if we assume the current to be anticlockwise then the magnetic field produced at the center of the loop will be upward which will assist the original field.

In Fig. 8.3.2(c), when the current is clockwise, the magnetic field at the center will be downward, opposing the original vertically up increasing field. According to Lenz's law this is the current direction of current.

The emf (V_{emf}) on the left side of Equation (8.3.3) is non zero when the right hand side is not zero. The non zero value of $\frac{d}{dt} \int \vec{B} \cdot d\vec{s}$ may result from any of the following situations :

1. A time changing flux and a path is stationary.
2. A time changing path and the magnetic field ($\vec{B}$ or ψ_m) is steady.
3. A combination of the two.

The first situation is observed in case of transformer so the situation is called as **transformer action**.

The second situation is observed in case of electric generator so it is called as **electric generator action**.

Thus the Faraday's law stated earlier is also expressed as :

Faraday's law states that the voltage around the closed path can be generated by a time changing magnetic flux through a fixed path (transformer action) or by a time changing path in a steady magnetic field (electric generator action).

As stated above emf induced is due to transformer action or electric generator action combined effect of two. This will give rise to three types of emfs.

- 1) Transformer emf : B changes with time and loop is stationary.
- 2) Motional emf : Here area change (otherwords loop moves) in a steady field.
- 3) Transformer and motional emf : Here moving loop is present in changing B.

8.3.2 Transformer emf :

In the section on Faraday's Law we learned that emf is induced in a loop when it is placed in the changing magnetic field. This is the principle on which transformer works like this -

In the transformer alternating current flowing in primary produces alternating flux which is confined to the core. The stationary turns cut this flux resulting in emf across each turn of the secondary. And we get voltage (V_2) across secondary. This emf is named as transformer emf.

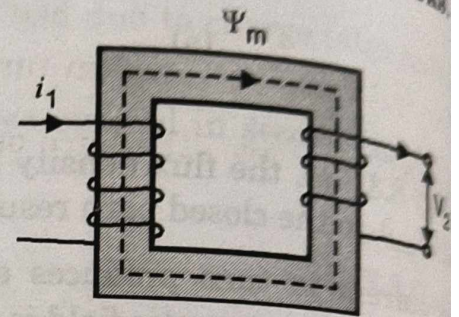


Fig. 8.3.3 : Transformer principle

Notice that transformer emf is due to stationary loop and changing magnetic field. Thus on the right side of Equation (8.3.3), can be altered by changing normal differentiation to partial differentiation.

$$V_{emf} = \oint \vec{E} \cdot d\vec{l} = - \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{s}$$

Assigning partial differentiation to only changing quantity (i.e. $\vec{B}$) we get,

$$V_{emf} = \oint \vec{E} \cdot d\vec{l} = - \int_S \frac{\partial}{\partial t} \vec{B} \cdot d\vec{s}$$

For simplicity in notation we also write it as

$$V_{emf} = \oint \vec{E} \cdot d\vec{l} = - \int_S \dot{\vec{B}} \cdot d\vec{s}$$

Where $\dot{\vec{B}}$ is the partial differential of $\vec{B}$ with respect to t.

Applying stoke's theorem to line integral

$$\oint \vec{E} \cdot d\vec{l} = \int_S (\nabla \times \vec{E}) \cdot d\vec{s} = - \int_S \dot{\vec{B}} \cdot d\vec{s}$$

or

$$\boxed{\nabla \times \vec{E} = - \dot{\vec{B}}}$$

8.3.3 Motional emf :

We know the generator principle. In the generator there are permanent magnets producing steady magnetic field. In the presence of this field a rotor attached to the loop is kept. When the rotor rotates, the loop rotates cutting the magnetic field. This results in emf induced in a loop. This emf is called motional emf. Notice the word motion in motional reminding us that the loop is in motion.

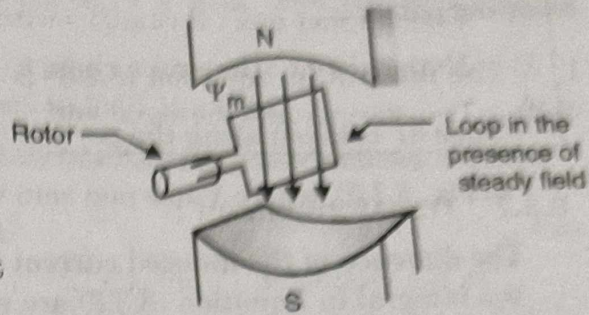


Fig. 8.3.4 : Generator principle

The emf in the loop will in turn gives rise to current in the loop, and then the loop will experience a force. The force on a charge moving with uniform velocity $\vec{V}$ in a magnetic field $\vec{B}$ is

$$\vec{F}_m = Q \vec{V} \times \vec{B}$$

The field obtained from it is called motional electric field ($\vec{E}_m$).

$$\vec{E}_m = \frac{\vec{F}_m}{Q} = \vec{V} \times \vec{B}$$

The current in the loop can be considered to be due to moving charges, so

$$V_{emf} = \oint \vec{E}_m \cdot d\vec{l} = \oint (\vec{V} \times \vec{B}) \cdot d\vec{l}$$

This type of emf is called as motional emf. or flux-cutting emf. Applying stoke's theorem to both integrals,

$$\int_S (\nabla \times \vec{E}_m) \cdot d\vec{s} = \int_S \nabla \times (\vec{V} \times \vec{B}) \cdot d\vec{s}$$

or

$$\nabla \times \vec{E}_m = \nabla \times (\vec{V} \times \vec{B})$$

Another example of the motional emf is shown in figure below. Here the conducting rod is moving between a pair of rails. The current is induced in the rod which in presence of field will exert force on the rod.

The magnetic field ($\vec{B}$) in the figure is perpendicular to the page (into) and the rod is moving in the plane of the paper. The two directions are perpendicular to each other. So the equation.

$$\vec{F}_m = Q \vec{V} \times \vec{B}$$

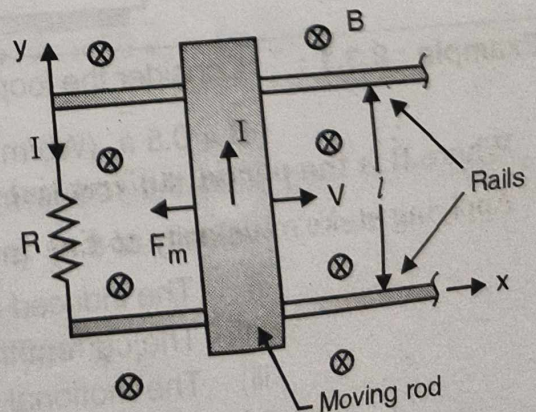


Fig. 8.3.5

reduces to $F_m = \bar{I} l \times \bar{B} \Rightarrow I l B$ and $V_{emf} = VB/l$

To apply Equation (8.3.3) is not always easy, some care must be exercised. Remember the following points :

1. The integral in Equation (8.3.3) is zero along the portion of the loop where $\bar{V} = 0$. Thus $d\bar{l}$ is taken along the portion of the loop that is cutting the field (along the rod in Fig. 8.3.5), where $\bar{V}$ has non zero value.
2. The direction of the induced current is the same as that of $\bar{E}_m$ or $\bar{V} \times \bar{B}$. The limits of the integral in Equation (8.3.3) are selected in the opposite direction to the induced current thereby satisfying Lenz's law. In Equation (8.3.3), for example, the integration over l is along $-\bar{a}_y$ where as induced current flows in the rod along $\bar{a}_y$.
- 3) **Moving Loop in time varying field :**
In this case the magnetic field is changing as well as the loop is moving. Then applying superposition,

$$V_{emf} = \oint \bar{E} \cdot d\bar{l} = - \int_S \dot{\bar{B}} \cdot d\bar{s} + \oint_l (\bar{V} \times \bar{B}) \cdot d\bar{l}$$

Applying Stoke's theorem to every line integral, gives

$$\int_S (\nabla \times \bar{E}) \cdot d\bar{s} = - \int_S \dot{\bar{B}} \cdot d\bar{s} + \int_S \nabla \times (\bar{V} \times \bar{B}) \cdot d\bar{s}$$

$$\boxed{\nabla \times \bar{E} = -\dot{\bar{B}} + \nabla \times (\bar{V} \times \bar{B})}$$

8.3.4 Solved Examples on Faraday's Law :

Important Formulae

$$\nabla \times \bar{E} = -\dot{\bar{B}}$$

$$\nabla \times \bar{E} = -\dot{\bar{B}} + \nabla \times (\bar{V} \times \bar{B})$$

Example : 8.3.1 : Consider the loop shown in Fig. Ex. 8.3.1. If $\bar{B} = 0.5 \bar{a}_z$ (Wb/m²), $R = 20 \Omega$, $l = 10$ cm, and the rod is moving with a constant velocity of $8 \bar{a}_x$ (m/s), find.

- i) The induced emf in the rod.
- ii) The current through the resistance.
- iii) The motional force on the rod.
- iv) The power dissipated by the resistor.

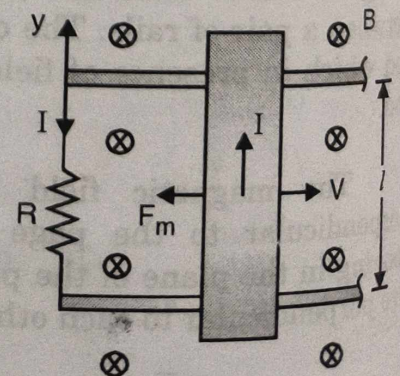


Fig. Ex. 8.3.1

i) To find the induced emf : Given, $\vec{B} = 0.5 \vec{a}_z$, $\vec{V} = 8 \vec{a}_z$ and the conductor is along y-axis,

$$V_{\text{emf}} = \int (\vec{V} \times \vec{B}) \cdot d\vec{l} = VB l = 8 (0.5) (0.1) = 0.4 \text{ V}$$

...Ans.

ii) To find the current :

$$I = \frac{V_{\text{emf}}}{R} = \frac{0.4}{20} = 20 \text{ (mA)}$$

...Ans.

iii) To find the motional force :

$$\vec{F}_m = I \vec{l} \times \vec{B} = 20 \times 10^{-3} (0.1 \vec{a}_y \times 0.5 \vec{a}_z) = -\vec{a}_x \text{ (mN)}$$

...Ans.

iv) The power dissipated by the resistor :

$$P = I^2 R = (20 \times 10^{-3})^2 \times 20 = 8 \text{ (mW)}$$

...Ans.

Example 8.3.2 : The loop shown in Fig. Ex. 8.3.2 is inside

a uniform magnetic field $\vec{B} = 50 \vec{a}_x \text{ (mWb/m}^2\text{)}$. If side DC of the loop cuts the flux lines at the frequency of 50 Hz and the loop lies in the yz-plane at time $t = 0$, find

- The induced emf at $t = 1 \text{ ms}$
- The induced current at $t = 3 \text{ ms}$.

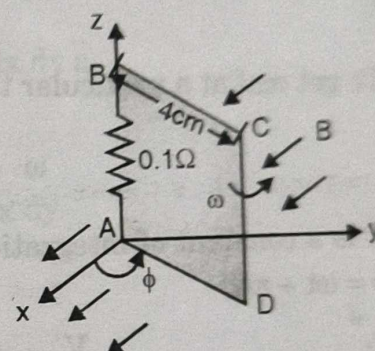


Fig. Ex. 8.3.2

Solution : In the example $\vec{B}$ is not changing with respect to time, so it is case 2 and the emf is motional emf.

$$V_{\text{emf}} = \int (\vec{V} \times \vec{B}) \cdot d\vec{l}$$

For side DC,

$$d\vec{l} = dz \vec{a}_z$$

The loop is rotating in ϕ direction giving

$$\vec{V} = \frac{r d\phi}{dt} \vec{a}_\phi \Rightarrow r \omega \vec{a}_\phi = 4 \times 10^{-2} \times 2\pi \times 50 \vec{a}_\phi = 4\pi \vec{a}_\phi$$

Where

$$\omega = d\phi / dt$$

We have $\vec{V}$ and $d\vec{l}$ in cylindrical so for calculation purpose convert $\vec{B}$ from cartesian to cylindrical. Let $\vec{B}$ in cylindrical be,

$$\vec{B} = B_r \vec{a}_r + B_\phi \vec{a}_\phi + B_z \vec{a}_z$$

where,

$$B_r = \vec{B} \cdot \vec{a}_r = 50 \vec{a}_x \times 10^{-3} \cdot \vec{a}_r = 0.05 \cos \phi$$

$$B_\phi = \vec{B} \cdot \vec{a}_\phi = 50 \vec{a}_x \times 10^{-3} \cdot \vec{a}_\phi = 0.05 \sin \phi$$

$$B_z = \vec{B} \cdot \vec{a}_z = 50 \vec{a}_z \times 10^{-3} \cdot \vec{a}_z = 0$$

Then $\vec{B}$ in cylindrical is

$$\vec{B} = 0.05 (\cos \phi \vec{a}_r - \sin \phi \vec{a}_\phi)$$

Calculation of emf requires cross product

$$\begin{aligned} \vec{V} \times \vec{B} &= \begin{bmatrix} \vec{a}_r & \vec{a}_\phi & \vec{a}_z \\ 0 & 4\pi & 0 \\ 0.05 \cos \phi & -0.05 \sin \phi & 0 \end{bmatrix} = -4\pi \times 0.05 \cos \phi \vec{a}_z \\ &= -0.63 \cos \phi \vec{a}_z \end{aligned}$$

$$(\vec{V} \times \vec{B}) \cdot d\vec{l} = -0.63 \cos \phi \vec{a}_z \cdot dz \vec{a}_z = -0.63 \cos \phi dz$$

$$V_{\text{emf}} = \int_0^{0.03} (\vec{V} \times \vec{B}) \cdot d\vec{l} = -0.63 \cos \phi \int_0^{0.03} dz = -18.9 \cos \phi \text{ (mV)}$$

To get emf at a particular time we should calculate ϕ at that time. We know

$$\omega = \frac{d\phi}{dt} \quad \text{or} \quad \phi = \omega t + C$$

C is a constant of integration. To calculate C, apply given condition at $t = 0$, $\phi = \pi/2$ given $\phi = \omega t + \pi/2$

$$V_{\text{emf}} = -18.9 \sin(\omega t + \pi/2) \Rightarrow -18.9 \sin(100\pi t) \text{ (mV)}$$

i) To calculate emf at 1 (ms),

$$V_{\text{emf}} = -18.9 \sin(100\pi \times 1 \times 10^{-3}) = 5.825 \text{ (mV)}$$

ii) To find current at $t = 3 \text{ ms}$

$$i = \frac{V_{\text{emf}}}{R} = 188.50 \sin(100\pi t) \text{ (mA)}$$

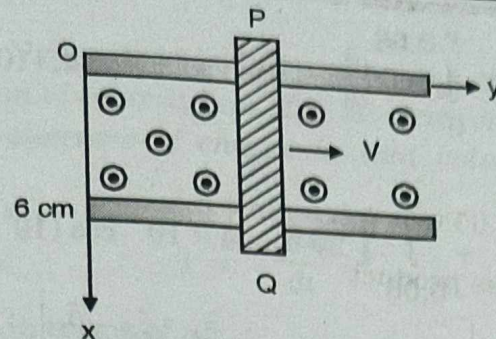
at $t = (3 \text{ ms})$,

$$\begin{aligned} i &= -188.50 \sin(100\pi \times 3 \times 10^{-3}) \text{ (mA)} \\ &= -188.50 \sin(0.3\pi) \text{ mA} = -0.1525 \text{ A} \end{aligned}$$

The magnitude is 0.1525 (A).

Example 8.3.3 : A conducting bar can slide freely over two conducting rails as shown in figure. Calculate the induced voltage in the bar.

- If the bar is stationary at $y = 8 \text{ cm}$ and $B = 4 \cos 10^6 t \vec{a}_z \text{ (mWb/m}^2\text{)}$.
- If the bar slides at the velocity $V = 20 \vec{a}_y \text{ (m/s)}$ and $\vec{B} = 4 \vec{a}_z \text{ (mWb/m}^2\text{)}$
- If the bar slides at a velocity $V = 20 \vec{a}_y \text{ (m/s)}$ and $\vec{B} = 4 \cos(10^6 t - y) \vec{a}_z \text{ (mWb/m}^2\text{)}$



Solution :

- (a) Here the bar is stationary so motional emf is absent, but $\vec{B} (= 4 \cos 10^6 t \cdot \vec{a}_z)$ is changing with respect to time, so transformer emf is present, given by,

$$\begin{aligned} V_{\text{emf}} &= - \int_S \dot{\vec{B}} \cdot d\vec{s} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \\ &= - \int \int -4 (\sin 10^6 t) \times 10^6 \times 10^{-3} \vec{a}_z \cdot dx dy \vec{a}_z \\ &= 4 \times 10^6 \times 10^{-3} \sin 10^6 t \int_{y=0}^{0.08} \int_{x=0}^{0.06} dx dy \\ &= 4 \times 10^6 \times 10^{-3} \times (x)_0^{0.06} (y)_0^{0.08} \sin 10^6 t \\ &= 19.2 \sin 10^6 t \text{ (Volts)} \end{aligned}$$

...Ans.

- (b) Here bar is moving so motional emf is present, but B is constant so transformer emf is absent. Thus the induced emf is,

$$\begin{aligned} V_{\text{emf}} &= \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} \\ &= \int_{x=0.06}^0 (20 \vec{a}_y \times 4 \vec{a}_z (10^{-3}) dx \vec{a}_x) = 80 \times 10^{-3} \int_{x=0.06}^0 \vec{a}_x \cdot dx \vec{a}_x \\ &= 80 \times 10^{-3} (x)_{0.06}^0 = -4.8 \text{ (mV)} \end{aligned}$$

...Ans.

- (c) Here both transformer emf and motional emf are present giving

$$V_{\text{emf}} = - \int \dot{\vec{B}} \cdot d\vec{s} + \oint (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

$$\begin{aligned}
&= - \int_0^y \int_0^{0.06} \frac{\partial}{\partial t} [4 \cos(10^6 t - y) \bar{a}_z (10^{-3})] \cdot dx dy \bar{a}_z \\
&\quad + \int_{0.06}^0 \left[20 \bar{a}_y \times 4 \times 10^{-3} \cos(10^6 t - y) \bar{a}_z \right] \cdot dx \bar{a}_x \\
&= + \int_0^y \int_0^{0.06} (4 \times 10^6 \times 10^{-3}) \sin(10^6 t - y) dx dy \\
&\quad + 20 \times 4 \times 10^{-3} \cos(10^6 t - y) \int_{0.06}^0 dx \\
&= 4 \times 10^6 \times 10^{-3} (x)_0^{0.06} [\cos(10^6 t - y)]_0^y + 20 \times 4 \times 10^{-3} \cos(10^6 t - y) (x)_{0.06}^0 \\
&= 240 \cos(10^6 t - y) - 240 \cos 10^6 t - 4.8 \times 10^{-3} \cos(10^6 t - y)
\end{aligned}$$

The last term is very small and can be neglected. Thus,

$$\begin{aligned}
V_{\text{emf}} &= 240 \cos(10^6 t - y) - 240 \cos(10^6 t) \\
&= 240 [\cos(10^6 t - y) - \cos(10^6 t)] \\
&= 240 \left[-2 \sin\left(10^6 t - \frac{y}{2}\right) \cdot \sin\left(\frac{y}{2}\right) \right] (\text{Volts})
\end{aligned}$$

Here the trigonometric identity,

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

is used. Thus,

$$V_{\text{emf}} = -480 \sin\left(10^6 t - \frac{y}{2}\right) \sin\left(\frac{y}{2}\right) (\text{Volts})$$

...Ans.